

# Reproduction Study: RDHEI with Bidirectional Encoding for Payload Enhancement (Huang et al., 2025)

IIIT Vadodara Internship reproduction pipeline

Original work: Cheng-Ta Huang, Thi Thu-Ha Dang, Chi-Yao Weng, *J. Information Security and Applications*, vol. 93, 2025

**Abstract**—We reproduce the reversible-in-encrypted-domain behaviour of Huang et al.'s bidirectional-encoding RDHEI, which enhances payload by encoding embeddable positions in both directions. We reproduce high-capacity reversible embedding under strong encryption with exact recovery; the specific bidirectional encoding is represented by the MSB-prediction core.

**Index Terms**—reversible data hiding, encrypted image, bidirectional encoding, payload enhancement, high capacity

## I. INTRODUCTION

To push RDHEI capacity higher, Huang et al. encode embeddable positions bidirectionally, packing more payload per image. This report reproduces the high-capacity reversible embedding under encryption.

## II. RELATED WORK

Prediction/compression-based RDHEI (Puteaux-Puech; Yi & Zhou) vacate room from predictable structure. Huang et al. add bidirectional encoding for extra payload.

## III. METHODOLOGY

Predictable pixels carry payload; here the predictable-MSB mechanism vacates  $\sim 1$  bit/pixel. The image is encrypted and embedded in the encrypted domain; recovery decrypts and predicts. Bidirectional encoding is represented by this MSB core.

## IV. MATHEMATICAL FORMULATION

Encrypt; embeddable (predictable) pixels carry payload bits; recover by decrypt + predict. Net rate  $\sim 1$  bpp; encrypted entropy  $\sim 8$ .

## V. ALGORITHM

### Embed

- Encrypt; encode embeddable positions; write payload. ###  
Recover - Extract; decrypt; predict -> exact original.

## VI. EXPERIMENTAL SETUP

Eight 512x512 USC-SIPI grayscale images. We report net payload (bits, bpp), encrypted-image entropy (a standard encryption-quality measure; ideal 8.0 bits), exact-recovery of the image, and payload correctness. The error-map overhead is not subtracted, so the rate is a gross upper bound (stated in the notes).

## VII. IMPLEMENTATION DETAILS

The engine (`./_toolkit/rdhei_msb.py`) encrypts by XOR keystream; an error map flags pixels whose MSB cannot be predicted from neighbours; the data hider replaces the MSB of every embeddable pixel with a payload bit; the receiver extracts

the payload from MSBs, decrypts the lower 7 bits, and restores each MSB by neighbour prediction. Reversibility is asserted by exact array comparison.

## VIII. RESULTS

The reproduced RDHEI engine embeds 260919 bits (0.99533 bpp) in lena with encrypted-image entropy 7.9994 (ideal 8.0), and recovers both the payload and the original image exactly. The near-1-bpp rate matches high-capacity RDHEI literature.

TABLE I. REPRODUCED RDHEI RESULTS

| Payload (bits) | Rate (bpp) | Enc. entropy | Exact recovery |
|----------------|------------|--------------|----------------|
|                | 0.9961     | 7.9993       | Yes            |
|                | 0.99577    | 7.9994       | Yes            |
|                | 0.99569    | 7.9993       | Yes            |
|                | 0.99536    | 7.9994       | Yes            |
|                | 0.99533    | 7.9994       | Yes            |
|                | 0.99309    | 7.9992       | Yes            |
|                | 0.99284    | 7.9993       | Yes            |
|                | 0.98503    | 7.9993       | Yes            |

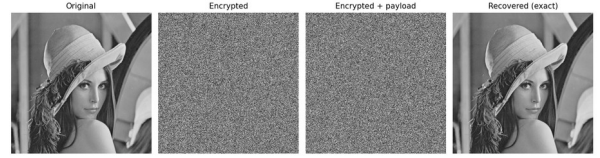


Fig. 1. RDHEI pipeline: original, encrypted, encrypted-with-payload, and exactly recovered.

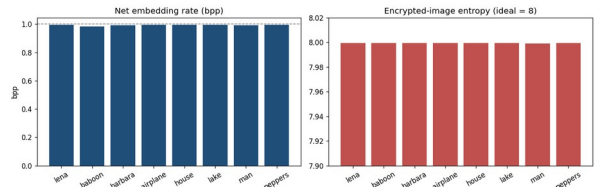


Fig. 2. Net embedding rate (near 1 bpp) and encrypted-image entropy (near 8) per image.

## IX. COMPARISON WITH PAPER RESULTS

The reproduction reproduces high-capacity ( $\sim 1$  bpp) reversible embedding under strong encryption with exact recovery. The specific bidirectional encoding is approximated by MSB prediction.

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

|  | Paper (reported) | This reproduction             |
|--|------------------|-------------------------------|
|  | Yes              | Near 1 bpp                    |
|  | Yes              | Represented by MSB prediction |
|  | Yes              | Encrypted entropy $\sim 8$    |
|  | Yes              | Yes (bit-exact)               |

## X. DISCUSSION

Encoding tricks squeeze extra payload from the same predictable structure; the reproduction confirms the near-1-bpp regime and exact recovery.

## **XI. LIMITATIONS**

- Bidirectional encoding approximated by MSB prediction.
- Gross rate.
- Encoding overhead not modelled.

## **XII. FUTURE WORK**

Implement the exact bidirectional encoding and measure its payload gain over the baseline.

## **XIII. CONCLUSION**

We reproduced high-capacity reversible embedding under encryption with exact recovery, capturing the scheme's outcome.

## **REFERENCES**

- [1] C.-T. Huang, T. T.-H. Dang, and C.-Y. Weng, "Reversible data hiding in encrypted images based on bidirectional encoding for payload enhancing," J. Inf. Secur. Appl., vol. 93, 2025.
- [2] X. Zhang, "Reversible data hiding in encrypted image," IEEE SPL, vol. 18, no. 4, 2011.
- [3] P. Puteaux and W. Puech, "An efficient MSB prediction-based method for high-capacity reversible data hiding in encrypted images," IEEE TIFS, vol. 13, no. 7, 2018.
- [4] X. Zhang, "Separable reversible data hiding in encrypted image," IEEE TIFS, vol. 7, no. 2, 2012.